

Assignment 2

1. Sector Identification

Pick one stock each of your choice from major sectors (Banking, Information Technology, Automobile, Pharmaceutical) in NSE/BSE.

Exchange Name (NSE/BSE)	Sector Name	Stock Name
	Banking	
	Information Technology	
	Automobile	
	Pharmaceutical	

2. Dataset Requirements

Obtain the Past 15 years of daily stock data. Use the following time split for training and testing trading strategies.

Dataset	Period	Purpose
Training	First 10 years	Strategy design
Testing	Last 5 years	Out-of-sample evaluation

3. Chart Observation

Use TradingView or any charting tool to visually inspect charts of the selected stocks.

- I. Identify Trend Patterns
 - a) Identify at least three major uptrends.
 - b) Identify two prolonged sideways markets.
 - c) What macroeconomic events correspond to these movements?
- II. Identify the Support and Resistance
 - a) Draw three support levels.
 - b) Draw three resistance levels.
 - c) Did price break these levels?
- III. Identify Chart Patterns (Head and Shoulders, Double Top, Double Bottom, Triangle)
 - a) Identify at least two chart patterns.
 - b) What happened after the breakout?
 - c) Did the pattern lead to a trend reversal?

4. Strategy Design and Evaluation

Design a strategy using maximum 4–5 indicators using the data of one of the stocks. Use the time split defined in point 2 above.

Metric	Training Data (First 10 Years)	Testing Data (Last 5 Years)
Total # of Trades		
Total Return		
Sharpe Ratio		
Maximum Drawdown		
Win Rate		

Example Strategy

Strategy: Trend + Momentum

Indicators used:

1. SMA50
2. SMA200
3. RSI
4. Volume

BUY when:

- SMA50 > SMA200
- RSI < 30
- Volume above average

SELL when:

- RSI > 70
- Price below SMA50

Submission Requirements

1. For Question 1 fill the sector table.
2. For Question 3 submit a short write-up and screenshots of charts clearly indicating observations.
3. For Question 4 submit the strategy code, filled table, and trade sheet.
4. For Question 5 submit filled tables and trade sheets for remaining three unseen stocks.